A characterization of cubic graphs with paired-domination number three-fifths their order

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Abstract

A paired-dominating set of a graph is a dominating set of vertices whose induced subgraph has a perfect matching, while the paired-domination number is the minimum cardinality of a paired-dominating set in the graph. Recently, Chen, Sun and Xing [Acta Mathematica Scientia Series A Chinese Edition 27(1) (2007), 166–170] proved that a cubic graph has paired-domination number at most three-fifths the number of vertices in the graph. In this paper, we show that the Petersen graph is the only connected cubic graph with paired-domination number three-fifths its order.

Keywords: bounds; cubic graphs; paired-domination; Petersen graph
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1 Introduction

A *matching* in a graph G is a set of independent edges in G. A *perfect matching* M is a matching such that every vertex of G is incident with an edge of M. A *paired-dominating set*, abbreviated PDS, of G is a set S of vertices of G such that every vertex is adjacent to some vertex in S and the subgraph $G[S]$ induced by S contains a perfect matching M (not necessarily induced). Every graph without isolated vertices has a PDS, since the end-vertices of any maximal matching form such a set. The *paired-domination number* of G,

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denoted by $\gamma_{\rm pr}(G)$, is the minimum cardinality of a PDS. A PDS of cardinality $\gamma_{\rm pr}(G)$ is a $\gamma_{\rm pr}(G)$-set. Paired-domination was introduced by Haynes and Slater [7, 8] as a model for assigning backups to guards for security purposes, and studied in [2, 3, 4, 5, 9] inter alia.

For notation and graph-theory terminology we in general follow [6]. Specifically, let $G = (V, E)$ be a graph with vertex set V of order $|V|$ and edge set E. For a set $S \subseteq V$, the subgraph induced by S is denoted by $G[S]$. The minimum degree among the vertices of G is denoted by $\delta(G)$. For disjoint subsets $A, B \subseteq V$, we define $[A, B]$ to be the set of all edges of G that join a vertex in A and a vertex in B. The *open neighborhood* of a vertex $v \in V$ is $N(v) = \{u \in V \mid uv \in E\}$, and the *closed neighborhood of* v is $N[v] = \{v\} \cup N(v)$. For a set $S \subseteq V$, its *open neighborhood* is the set $N(S) = \cup_{v \in S} N(v)$ and its *closed neighborhood* is the set $N[S] = N(S) \cup S$. If X and Y are subsets of vertices in G, then the set X dominates Y in G if $Y \subseteq N[X]$.

1.1 Known Results

The decision problem to determine the paired-domination number of a graph is known to be NP-complete ([7]). Hence it is of interest to determine upper bounds on the paired-domination number of a graph. Haynes and Slater [7] obtained the following upper bound on the paired-domination number of a connected graph in terms of the order of the graph.

Theorem 1 (Haynes, Slater [7]) *If G is a connected graph of order $n \geq 3$, then $\gamma_{\rm pr}(G) \leq n - 1$ with equality if and only if G is C_3, C_5 or a subdivided star.*

If we restrict the minimum degree to be at least two and the order to be at least six, then the upper bound in Theorem 1 on the paired-domination number can be improved from one less than its order to two-thirds its order.

Theorem 2 (Haynes, Slater [7]) *If G is a connected graph of order $n \geq 6$ with $\delta(G) \geq 2$, then $\gamma_{\rm pr}(G) \leq 2n/3$.*

The upper bound in Theorem 2 can be improved slightly if the order of the graph is large.

Theorem 3 ([9]) *If G is a connected graph of order $n \geq 10$ with $\delta(G) \geq 2$, then $\gamma_{\rm pr}(G) \leq 2(n - 1)/3$ and there are infinite families of graphs that achieve equality in this bound.*

A characterization of connected graphs with minimum degree at least two and order at least fourteen that achieve the upper bound in Theorem 3 is given in [9]. Recently, Chen, Sun and Xing [1] proved the following result.

Theorem 4 ([1]) *If G is a cubic graph of order n, then $\gamma_{\rm pr}(G) \leq 3n/5$.*

Although the sharpness of the bound in Theorem 4 is not discussed in [1], we remark that the Petersen graph P shown in Figure 1 of order $n = 10$ with $\gamma_{\rm pr}(P) = 6$ is a cubic graph that achieves equality in the bound of Theorem 4.

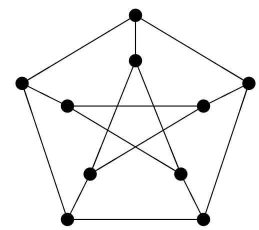

Figure 1: The Petersen Graph P.

2 Main Result

Our aim in this paper is twofold: First, to provide a slightly simpler proof of Theorem 4, and second, to use this new proof to characterize the cubic graphs that achieve equality in the bound of Theorem 4. We shall prove:

Theorem 5 *If G is a connected cubic graph of order n, then $\gamma_{\rm pr}(G) \le 3n/5$ with equality if and only if G is the Petersen graph.*

We shall proceed as follows. In Section 2.1, we choose a minimum PDS S that induces a subgraph with the minimum number of edges, and fix a perfect matching in $G[S]$ that defines our pairing of the vertices in S. Basic properties of the PDS S are then discussed. In Section 2.2, we assign weights to the resulting pairs of vertices in S. In Section 2.3, we show that every such pair has weight at least $4/3$. From this we deduce that S contains at most three-fifths the vertices. In Section 2.4, we show that the Petersen graph is the unique connected cubic graph with paired-domination number exactly three-fifths its order.

2.1 The Paired-Dominating Set S

Let $G = (V, E)$ be a connected cubic graph of order n. For a subset $S \subseteq V$, let $\lambda(S)$ be the number of edges in $G[S]$.

Among all minimum PDS of G, let S be chosen so that $\lambda(S)$ is minimized.

Necessarily, S is a minimal PDS of G; that is, no proper subset of S is a PDS of G.

Fix some perfect matching M in $G[S]$. For each vertex $v \in S$, let $\overline{v}$ denote the vertex such that $v\overline{v} \in M$. We will refer to $\overline{v}$ as the *partner* of v. Let $S_v = \{v, \overline{v}\}$; we will refer to S_v as a *pair in S*.

Let $v \in S$. Then the *S-private neighborhood* of v is the set $\mathrm{pn}(v,S) = N[v] \setminus N[S \setminus \{v\}]$. We call a vertex $u \in \mathrm{pn}(v,S)$ an *S-private neighbor* of v. If $S' \subseteq S$, then the *private neighborhood set* of S' is the set consisting of all S-private neighbors of vertices in S'; that is, $\mathrm{pn}(S',S) = \bigcup_{v \in S'} \mathrm{pn}(v,S)$.

We define a weak partition (A,B,C,D) of the set S (where by *weak partition* we mean that some of the subsets may be empty) as follows. Let A consist of all vertices of S for which the sum of the number of S-private neighbors of the vertex and its partner is at least 2. Let B consist of all vertices of S that have exactly one S-private neighbor and whose partner has no S-private neighbor. Let C consist of all vertices of S with no S-private neighbor and whose partner belongs to B. Let D consist of all vertices of S with no S-private neighbor and whose partner does not belong to B (and therefore has no S-private neighbor). That is,

$$\begin{array}{rcl} A & = & \{\, v \in S : |\mathrm{pn}(S_v,S)| \geq 2 \,\} \\ B & = & \{\, v \in S : |\mathrm{pn}(S_v,S)| = |\mathrm{pn}(v,S)| = 1 \,\} \\ C & = & \{\, v \in S : \overline{v} \in B \,\} \\ D & = & \{\, v \in S : |\mathrm{pn}(S_v,S)| = 0 \,\}. \end{array}$$

We define a pair of vertices in S (that are partners) to be:

an A-pair if both belong to A;

a BC-pair if one belongs to B and the other to C; and

a D-pair if both belong to D.

We remark that every pair in S form either an A-pair or a BC-pair or a D-pair.

We proceed further with a series of claims.

Claim A *Every vertex in C has two neighbors outside S.*

Proof. Let $v \in C$ and suppose that vertex v is adjacent to a vertex of S other than $\overline{v}$. Then replacing v in S by the S-private neighbor of its partner $\overline{v}$ produces a new PDS S' with $\lambda(S') < \lambda(S)$, contradicting our choice of S QED

Claim B *At least one vertex from every D-pair has two neighbors outside S.*

Proof. Consider a D-pair $\{v,\overline{v}\}$. Assume that neither v nor $\overline{v}$ has two neighbors outside S. Then, both v and $\overline{v}$ have at least two neighbors in $G[S]$. By the minimality of S, the set $S \setminus \{v,\overline{v}\}$ does not dominate V. But $S \setminus \{v,\overline{v}\}$ dominates v and $\overline{v}$, and neither v nor $\overline{v}$ has an S-private neighbor. Hence, there must be a vertex w of $V \setminus S$ whose neighbors in S are precisely v and $\overline{v}$. But then, replacing the vertex v in S by w produces a new PDS S' with $\lambda(S') < \lambda(S)$, a contradiction. Hence at least one of v and $\overline{v}$ has two neighbors outside S. QED

Claim C *If there is an edge joining vertices from two distinct D-pairs, then all other edges incident with these four vertices belong to $[S, V \setminus S]$.*

Proof. Assume that there is an edge $uv \in E$ where $\{u, \overline{u}\}$ and $\{v, \overline{v}\}$ are two distinct D-pairs. Consider the set $S_1 = S \setminus \{\overline{u}, \overline{v}\}$. By the minimality of S, the set S_1 does not dominate V. But S_1 dominates $\overline{u}$ and $\overline{v}$, and neither $\overline{u}$ nor $\overline{v}$ has an S-private neighbor. Hence, there must be a vertex $w \in V \setminus S$ whose neighbors in S are precisely $\overline{u}$ and $\overline{v}$. But then consider the PDS S_2 obtained from S by replacing the vertex u by w (with $\overline{u}$ paired with w). If u has no neighbor outside S, then $\lambda(S_2) < \lambda(S)$, a contradiction. Hence the neighbor of u different from $\overline{u}$ and v is outside S. Similarly, the neighbor of v different from u and $\overline{v}$ is outside S. By Claim B, each of $\overline{u}$ and $\overline{v}$ has two neighbors outside S. QED

Assume that there is an edge $uv \in E$ where $S_u = \{u, \overline{u}\}$ and $S_v = \{v, \overline{v}\}$ are distinct D-pairs. By Claim C, all other edges incident with these four vertices belong to $[S, V \setminus S]$. Therefore every D-pair that is joined to another D-pair is joined to a unique such D-pair. We define a D-pair to be

a *linked D-pair* if it is joined by an edge to some other D-pair; and

a *solo D-pair* if it is not a linked D-pair.

See Figure 2.

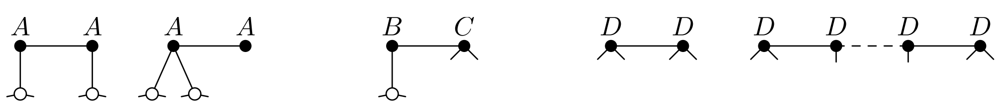

Figure 2: S consists of A-pairs, BC-pairs and D-pairs.

2.2 Defining the Weights

The general strategy is to define a weight on all the edges that join S to $V \setminus S$. This weight is defined so that for each vertex in $V \setminus S$, the total weight of the edges incident with it sums to 1. Thus the total weight is exactly $|V \setminus S|$. At the same time, we sum the weights for each pair in S, and after a suitable redistribution, we show that each pair has associated with it a weight of at least $4/3$. It follows that the total weight is at least $2|S|/3$, and by algebra it follows immediately that $|S| \le 3n/5$. In this regard, our approach is similar to what is seemingly followed in the original proof. Equality then requires that the weight associated with each pair is exactly $4/3$, and from that we deduce that the graph must be the Petersen graph.

The simplest idea for a weight function is that, for each vertex x in $V \setminus S$, weight 1 is shared among the one, two or three edges that join x to S. Under this scenario, using the above claims, it follows readily that an A-pair has weight at least 2 incident with it, and a BC-pair weight at least $5/3$, but D-pairs might only have weight $2/3$ if they have edges to other vertices in S. So the next step is to redistribute the excess from A- and B-pairs to boost the weight of D-pairs. Unfortunately, this still leaves linked D-pairs with insufficient weight; so one has to go back and produce a more complicated weight function.

So, we next define the sets X, Y and Z as follows:

$$\begin{array}{rcl} X & = & \{v \in V \setminus S \mid v \text{ has a neighbor in both } A \cup B \text{ and } D\} \\ Y & = & \{v \in V \setminus S \mid v \text{ has two neighbors in } C \text{ and one neighbor in } D\} \\ Z & = & \{v \in V \setminus S \mid v \text{ has one neighbor in } C \text{ and two neighbors in } D\}. \end{array}$$

We define a vertex to be an X-vertex if it belongs to X, a Y-vertex if it belongs to Y, and a Z-vertex if it belongs to Z.

We now define a function $\omega\colon [S, V \setminus S] \to [0,1]$ that assigns to each edge in $[S, V \setminus S]$ a weight. For each vertex $x \in V \setminus S$, the function ω assigns weights to the edges that join x to S as follows:

1. If x is an S-private neighbor, then assign the weight 1 to the unique edge from x to S.
2. If x is an X-vertex, then assign the weight 0 to the edge(s) from x to $A \cup B$ and share the weight 1 equally among the (one or two) edge(s) from x to $C \cup D$.
3. If x is a Y-vertex, then assign the weight $1/6$ to the two edges from x to C and assign the weight $2/3$ to the edge from x to D.
4. If x is a Z-vertex, then assign the weight $1/6$ to the edge from x to C and assign the weight $5/12$ to the two edges from x to D.
5. If x is neither an S-private neighbor nor an X-, Y- or Z-vertex, then share the weight 1 equally among the (two or three) edges from x to S.

The following claim follows readily from the definition of the function ω.

Claim D *Let $x \in V \setminus S$ and let the edge e join x to S. Then the following properties hold.*
(a) $\omega(e) \in \{0, \frac{1}{6}, \frac{1}{3}, \frac{5}{12}, \frac{1}{2}, \frac{2}{3}, 1\}$.
(b) The sum of the weights assigned to the edges that join x to S is 1.
(c) Assume e joins x to C. Then $\omega(e) \geq 1/6$, with equality only when x is a Y-vertex or a Z-vertex.
(d) Assume e joins x to D. Then $\omega(e) \geq 1/3$. If $\omega(e) = 1/3$, then there are three edges from x to D. If $\omega(e) = 5/12$, then x is a Z-vertex

We next define a function f that assigns to each subset $S' \subseteq S$ the sum of the weights of the edges from S' to $V \setminus S$; that is,

$$f(S') = \sum_{e \in [S', V \setminus S]} \omega(e).$$

If $S' = \{v\}$, we denote $f(S')$ simply by $f(v)$ rather than $f(\{v\})$. If $S' = S$, then $f(S)$ is the sum of the weights of all edges in $[S, V \setminus S]$ (namely, $|V \setminus S|$).

Finally, we define a function g that assigns to each pair $S_v = \{v, \overline{v}\}$ in S the weight

$$g(S_v) = \begin{cases} f(S_v) - \frac{1}{3}|[S_v, D]| & \text{if } v \in A \cup B \cup C \\ f(S_v) + \frac{1}{3}|[S_v, A \cup B]| & \text{if } v \in D. \end{cases}$$

Thus if $v \in A \cup B \cup C$, then the function g assigns to the pair S_v the weight $f(S_v)$ minus 1/3 for every edge joining S_v and D. And, if $v \in D$, then the function g assigns to the pair S_v the weight $f(S_v)$ plus 1/3 for every edge joining S_v and $A \cup B$ (by Claim A there is no edge joining v to C).

2.3 The Weight of each Pair

We consider the four different types of pairs, namely an A-pair, a BC-pair, a solo D-pair, and a linked D-pair, in turn. We show that each pair has weight at least 4/3.

Claim E *If S_v is an A-pair, then $g(S_v) \ge 4/3$.*

Proof. The A-pair S_v has at least two S-private neighbors, and so $f(S_v) \ge 2$. Therefore since $|[S_v, D]| \le 2$, we have that $g(S_v) \ge f(S_v) - 2/3 \ge 4/3$. QED

Claim F *If S_v is a BC-pair, then $g(S_v) \ge 4/3$.*

Proof. We may assume that $v \in B$ and $\overline{v} \in C$. Since v has an S-private neighbor, $f(v) \ge 1$. By Claim A, there are two edges that join $\overline{v}$ to $V \setminus S$. By Claim D(c), each such edge has weight at least 1/6, and so $f(\overline{v}) \ge 1/3$. Hence, $f(S_v) = f(v) + f(\overline{v}) \ge 4/3$. If $[S_v, D] = \emptyset$, then $g(S_v) = f(S_v) \ge 4/3$, as desired.

Hence we may assume that there is (exactly) one edge joining v to D, say vu. Consider the set $S_1 = S \setminus \{\overline{u}, \overline{v}\}$. By the minimality of S, the set S_1 does not dominate V. But S_1 dominates $\overline{u}$ and $\overline{v}$, and neither $\overline{u}$ nor $\overline{v}$ has an S-private neighbor. Hence there must be a vertex $x \in V \setminus S$ whose neighbors in S are precisely $\overline{u}$ and $\overline{v}$. By definition of the function ω, the edge joining $\overline{v}$ and x has weight 1/2. The remaining edge from $\overline{v}$ to $V \setminus S$ has weight at least 1/6, and so $f(S_v) = f(v) + f(\overline{v}) \ge 1 + 1/2 + 1/6 = 5/3$. Therefore since $|[S_v, D]| = 1$, we have that $g(S_v) = f(S_v) - 1/3 \ge 4/3$. QED

Claim G *If S_v is a solo D-pair, then $g(S_v) \ge 4/3$.*

Proof. Since there is no edge from S_v to $D \setminus S_v$, there are four edges from S_v to $V \setminus D$. Let e be any such edge. Then either $e \in [S_v, V \setminus S]$ or $e \in [S_v, A \cup B]$. If the former, then by

Claim D(d), $\omega(e) \geq 1/3$. Hence, $g(S_v) = f(S_v) + \frac{1}{3}|[S_v, A \cup B]| \geq \frac{1}{3}|[S_v, V \setminus S]| + \frac{1}{3}|[S_v, A \cup B]| = \frac{1}{3}|[S_v, V \setminus D]| = 4/3$. QED

The case of a linked D-pair is harder.

Claim H *If S_v is a linked D-pair, then $g(S_v) \geq 4/3$.*

PROOF. By definition, there is an edge that joins S_v to another D-pair, say $S_u = \{u, \overline{u}\}$. We may assume that $uv \in E$. Consider the set $S_1 = S \setminus \{\overline{u}, \overline{v}\}$. By the minimality of S, the set S_1 does not dominate V. But S_1 dominates $\overline{u}$ and $\overline{v}$, and neither $\overline{u}$ nor $\overline{v}$ has an S-private neighbor. Hence there must be a vertex $y \in V \setminus S$ whose neighbors in S are precisely $\overline{u}$ and $\overline{v}$. By definition of ω, the edges $y\overline{u}$ and $y\overline{v}$ both have weight $1/2$. By Claim C, the four edges incident with u, $\overline{u}$, v, $\overline{v}$ not on the cycle $y\,\overline{u}\,u\,v\,\overline{v}\,y$ belong to $[D, V \setminus S]$. By Claim D(d), each such edge has weight at least $1/3$. Let x be the neighbor of $\overline{v}$ different from v and y, and let z be the neighbor of v different from $\overline{v}$ and y (possibly, $x = z$). See Figure 3.

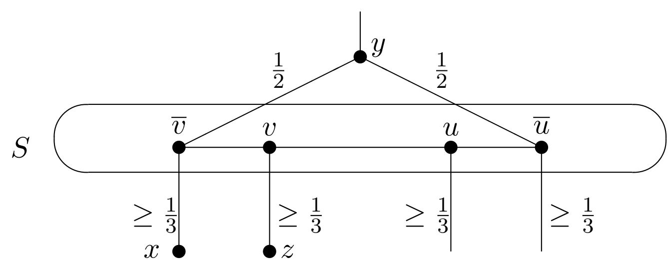

Figure 3: The edges incident with vertices in $S_u \cup S_v$.

If $\omega(\overline{v}x) \geq 1/2$ or $\omega(vz) \geq 1/2$, then $g(S_v) = f(S_v) = \omega(vz) + \omega(\overline{v}x) + \omega(\overline{v}y) \geq 4/3$, and we are done. Hence we may assume that $\omega(\overline{v}x), \omega(vz) < 1/2$. Thus, by Claim D(d),

each of x and z either has three neighbors in D or is a Z-vertex.

Suppose that all three neighbors of x are in $S_u \cup S_v$. There are three possibilities. If $N(x) = \{u, v, \overline{v}\}$, let $S' = (S \setminus \{u, \overline{v}\}) \cup \{x, y\}$. Then, S' is a PDS of G (with v and x paired and $\overline{u}$ and y paired in S') with $\lambda(S') < \lambda(S)$, contradicting the choice of S. If $N(x) = \{\overline{u}, v, \overline{v}\}$, let $S' = (S \setminus \{u, v, \overline{v}\}) \cup \{x\}$. Then, S' is a smaller PDS of G (with $\overline{u}$ and x paired), a contradiction. If $N(x) = \{u, \overline{u}, \overline{v}\}$, let $S' = (S \setminus \{u, v, \overline{u}\}) \cup \{x\}$. Then, S' is a smaller PDS of G (with $\overline{v}$ and x paired), a contradiction.

Suppose some neighbor of x, say w, is in $D \setminus (S_u \cup S_v)$. Let $S' = (S \setminus \{v, \overline{v}, \overline{w}\}) \cup \{x\}$. If S' dominates V, then S' is a smaller PDS of G (with x and w paired), a contradiction. Hence, S' does not dominate V. Therefore there must be a vertex $x' \in V \setminus S$ whose neighbors in S are precisely v and $\overline{w}$. Since z is the only neighbor of v not in $S_u \cup S_v$, we have that $x' = z$. Thus, z has exactly two neighbors in S, contradicting our earlier deduction.

It follows that x has a neighbor outside D. Thus x is a Z-vertex. Indeed, x has one neighbor in C, say w, and two neighbors in $S_u \cup S_v$. Also, $\omega(\overline{v}x) = 5/12$.

If $\omega(vz) \geq 5/12$, then $g(S_v) = \omega(\overline{v}x) + \omega(vz) + \omega(\overline{v}y) \geq 4/3$, as desired. So assume that $\omega(vz) = 1/3$. Then by the previous discussion, there are three edges from z to D. In particular, we note that $x \neq z$.

Suppose that all three neighbors of z are in $S_u \cup S_v$. Then, $N(z) = \{u, \overline{u}, v\}$. Let $S' = (S \setminus \{v, \overline{u}\}) \cup \{y, z\}$. Then, S' is a PDS of G (with u and z paired and y and $\overline{v}$ paired) with $\lambda(S') < \lambda(S)$, a contradiction. Hence at least one neighbor of z is not in $S_u \cup S_v$, say t.

We know that $t \in D$. Let $S' = (S \setminus \{u, v, \overline{v}, \overline{t}\}) \cup \{y, z\}$. If S' dominates V, then S' would be a smaller PDS of G (with $\overline{u}$ and y paired and t and z paired), a contradiction. Hence, S' does not dominate V. Note that since $w \in S'$, the vertex x is dominated by S'. Therefore, there must be a vertex $s \in V \setminus S$ whose neighbors in S are precisely u and $\overline{t}$.

From our earlier observations, the three vertices x, z and s are all distinct (x is a Z-vertex, z has three neighbors in D, while s has two neighbors in S). Hence, the neighbor of x in $S_u \cup S_v$ different from $\overline{v}$ must be the vertex $\overline{u}$. See Figure 4.

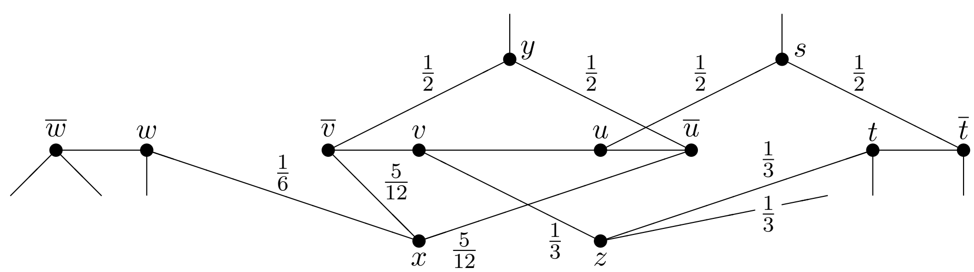

Figure 4: The edges incident with vertices in $S_u \cup S_v$.

Finally, we let $p \in D$ be the neighbor of z different from t and v. Let $S' = (S \setminus \{u, v, \overline{v}, \overline{p}\}) \cup \{y, z\}$. Then, S' is a smaller PDS of G (with $\overline{u}$ and y paired and p and z paired), a contradiction. QED

Thus we have shown that every pair S_v in S has weight $g(S_v) \geq 4/3$.

Claim I $|S| \leq 3n/5$.

Proof. Let S_P be a subset of S that consists of one vertex from each pair in S. By the above four claims, $g(S_v) \geq 4/3$ for every $v \in S_P$. Hence,

$$\sum_{v \in S_P} g(S_v) \geq \frac{4}{3}|S_P| = \frac{2}{3}|S|.$$

However, since

$$\bigcup_{v \in A \cup B \cup C} [S_v, D] = [A \cup B, D] = \bigcup_{v \in D} [A \cup B, S_v],$$

we have that

$$\sum_{v \in S_P} g(S_v) = \sum_{v \in S_P} f(S_v) - \frac{1}{3}[A \cup B, D] + \frac{1}{3}[A \cup B, D] = f(S) = n - |S|.$$

Hence, $2|S|/3 \leq n - |S|$, or, equivalently, $|S| \leq 3n/5$. QED

This establishes the upper bound of Theorem 5.

2.4 Equality implies the Petersen Graph

Let G be a connected cubic graph of order n that achieves equality in the bound of Theorem 5. That is, $\gamma_{\rm pr}(G) = 3n/5$. We show that G is the Petersen Graph P shown in Figure 1. We proceed further with a series of claims that we may assume the graph G to satisfy.

Claim J *The graph G has the following properties.*
(a) *Every pair S_v in S has weight $g(S_v) = 4/3$.*
(b) *Every edge from a solo D-pair to a vertex outside S has weight $1/3$.*
(c) *If uv is an edge that joins two linked D-pairs S_u and S_v, then there is a vertex in $V \setminus S$ whose neighbors in S are precisely $\overline{u}$ and $\overline{v}$.*

Proof. (a) Since $|S| = 3n/5$, we have equality throughout the proof of Claim J. In particular, every pair S_v in S has weight $g(S_v) = 4/3$.

(b) Let S_v be a solo D-pair. If there is an edge $e \in [S_v, V \setminus S]$ of weight exceeding $1/3$, then the proof of Claim G shows that $g(S_v) > 4/3$, contradicting (a).

(c) Consider the set $S_1 = S \setminus \{\overline{u}, \overline{v}\}$. By the minimality of S, the set S_1 does not dominate V. Hence there must be a vertex in $V \setminus S$ whose neighbors in S are precisely $\overline{u}$ and $\overline{v}$. QED

Claim K *There is no A-pair.*

Proof. Suppose that S_v is an A-pair. Since $g(S_v) = 4/3$, the proof of Claim E shows that there are exactly two edges from S_v to D, that is, $|[S_v, D]| = 2$. If both these edges are incident with exactly one of v and $\overline{v}$, say with v, then replacing v in S by one of the two S-private neighbors of its partner $\overline{v}$ produces a new PDS S' with $\lambda(S') < \lambda(S)$, a contradiction. Hence, one of the edges from S_v to D is incident with v and the other with $\overline{v}$.

Suppose the two edges from S_v to D go to different D-pairs. Then there exist two vertices u and w in D such that $S_u \neq S_w$ and $\{uv, \overline{v}w\} \subset E$. By Claim C, both S_u and S_w are solo D-pairs. Let $S' = S \setminus \{\overline{u}, \overline{w}\}$. If S' dominates V, then S' would be a smaller PDS of G (with u and v paired and $\overline{v}$ and w paired), a contradiction. Hence, S' does not dominate V. Therefore there must be a vertex $t \in V \setminus S$ whose neighbors in S are precisely

$\overline{u}$ and $\overline{w}$. By definition of the function ω, the two edges joining t to D both have weight $1/2$, contradicting Claim J(b). Hence the two edges from S_v to D go to the same D-pair.

By Claim B, at least one vertex from every D-pair has two neighbors outside S, and so the two edges from S_v to D are incident with a common vertex from this D-pair, say u. Further, by Claim B, the partner $\overline{u}$ of u has two neighbors outside S, say x and y. By Claim C, S_u is a solo D-pair. By Claim J(b), the two edges $\overline{u}x$ and $\overline{u}y$ both have weight $1/3$. Hence by Claim D(d), there are three edges from each of x and y to D. Let $w \in D$ be a neighbor of x different from $\overline{u}$. Let $S' = (S \setminus \{u, \overline{u}, \overline{w}\}) \cup \{x\}$. Since y is triply dominated by S while every neighbor of $\overline{w}$ outside S is doubly dominated by S, the set S' is a smaller PDS of G (with x and w paired), a contradiction. QED

Claim L *There is no Y-vertex.*

Proof. Suppose there is a Y-vertex y. By definition, the vertex y has two neighbors in C and a neighbor v in D. By definition of ω, $\omega(vy) = 2/3$. By Claim J(b), S_v is not a solo D-pair. Thus, S_v is a linked D-pair, and so there is an edge that joins S_v to some other D-pair, say S_u. By Claim J(c), there is a vertex t outside S that is adjacent to exactly one vertex of S_v and to exactly one vertex of S_u, but to no other vertex of S. By definition of the function ω, the edge that joins t to S_v has weight $1/2$. By Claim D(d), any edge from S_v to a vertex outside S has weight at least $1/3$. Hence, $g(S_v) \geq 2/3 + 1/2 + 1/3 > 4/3$, contradicting Claim J(a). QED

Claim M *There is no X-vertex.*

Proof. Suppose there is an X-vertex x. By Claim K, there is no A-pair. Hence by definition, the vertex x has a neighbor $w \in B$ and a neighbor $v \in D$. By Claim D(d), $\omega(vx) \geq 1/2$. Hence by Claim J(b), S_v is not a solo D-pair. Thus, S_v is a linked D-pair, and so there is an edge that joins S_v to some other D-pair, say $S_u = \{u, \overline{u}\}$. We may assume that either $uv \in E$ or $u\overline{v} \in E$.

Case 1: $uv \in E$. By Claim J(c), there is a vertex $y \in V \setminus S$ whose neighbors in S are precisely $\overline{u}$ and $\overline{v}$. By definition of ω, the edges $y\overline{u}$ and $y\overline{v}$ both have weight $1/2$. Let z be the neighbor of $\overline{v}$ different from v and y (possibly, $x = z$). Then, $\omega(\overline{v}z) \geq 1/3$. Since $g(S_v) = \omega(vx) + \omega(\overline{v}y) + \omega(\overline{v}z) = 4/3$, it follows that $\omega(vx) = \omega(\overline{v}y) = 1/2$ and $\omega(\overline{v}z) = 1/3$.

By Claim D(d), there are three edges from z to D. In particular, we note that $x \neq z$. If all three neighbors of z are in $S_u \cup S_v$, then $N(z) = \{u, \overline{u}, \overline{v}\}$ and $S' = (S \setminus \{u, \overline{u}, v\}) \cup \{z\}$ is a smaller PDS of G (with $\overline{v}$ and z paired), a contradiction. Hence at least one neighbor, t say, of z is not in $S_u \cup S_v$. As observed earlier, $t \in D$. Let $S' = (S \setminus \{v, \overline{v}, \overline{t}\}) \cup \{z\}$. Then, S' is a smaller PDS of G (with t and z paired), a contradiction.

Case 2: $u\overline{v} \in E$. Consider the set $S_1 = S \setminus \{\overline{u}, v\}$. By the minimality of S, the set S_1 does not dominate V. Hence there must be a vertex $y \in V \setminus S$ whose neighbors in S are precisely $\overline{u}$ and v. By definition of ω, the edges $y\overline{u}$ and yv both have weight 1/2. Let z be the neighbor of $\overline{v}$ different from u and v (possibly, $x = z$). Then, $\omega(\overline{v}z) \geq 1/3$. Since $g(S_v) = \omega(vx)+\omega(vy)+\omega(\overline{v}z) = 4/3$, it follows that $\omega(vx) = \omega(vy) = 1/2$ and $\omega(\overline{v}z) = 1/3$. By Claim D(d), there are three edges from z to D. In particular, we note that $x \neq z$.

Suppose that all three neighbors of z are in $S_u \cup S_v$. Then, $N(z) = \{u, \overline{u}, \overline{v}\}$. Let $S' = (S \setminus \{\overline{u}, \overline{v}\}) \cup \{y, z\}$. Then, S' is a PDS of G (with u and z paired and v and y paired) with $\lambda(S') < \lambda(S)$, a contradiction. Hence at least one neighbor of z is not in $S_u \cup S_v$.

Let t be a neighbor of z not in $S_u \cup S_v$. As observed earlier, $t \in D$. Let $S' = (S \setminus \{u, v, \overline{v}, \overline{t}\}) \cup \{y, z\}$. If S' dominates V, then S' would be a smaller PDS of G (with $\overline{u}$ and y paired and t and z paired), a contradiction. Hence, S' does not dominate V. Note that since there is an edge from x to B, the vertex x is dominated by S'. Therefore there must be a vertex $s \in V \setminus S$ whose neighbors in S are precisely u and $\overline{t}$. Note that the three vertices x, z and s are all distinct (x is an X-vertex, z has three neighbors in D, while s has two neighbors in S).

Let p be the neighbor of z distinct from $\overline{v}$ and t. As observed earlier, $p \in D$. Suppose that $p = \overline{u}$. The edges incident with vertices in $S_u \cup S_v$ are illustrated in Figure 5. Let $S' = (S \setminus \{v, \overline{v}, \overline{w}\}) \cup \{x\}$. Then S' is a smaller PDS of G (with w and x paired), a contradiction. Hence, $p \neq \overline{u}$. But then $S' = (S \setminus \{u, v, \overline{v}, \overline{p}\}) \cup \{y, z\}$ is a smaller PDS of G (with $\overline{u}$ and y paired and p and z paired), a contradiction. We deduce, therefore, that there is no X-vertex. QED

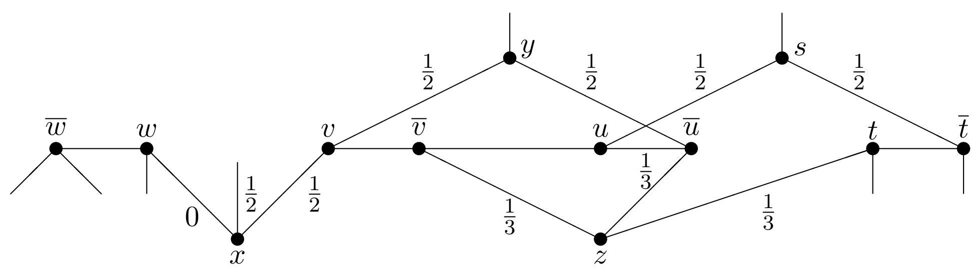

Figure 5: The edges incident with vertices in $S_u \cup S_v$.

Claim N *There is no BC-pair.*

Proof. Suppose that S_v is a BC-pair. We may assume that $v \in B$, and so $\overline{v} \in C$. Suppose there is an edge uv joining v to a vertex $u \in D$. By Claim C, the pair S_u is a solo D-pair. As shown in the proof of Claim F, there must be a vertex $x \in V \setminus S$ whose neighbors in S are precisely $\overline{u}$ and $\overline{v}$. By definition of ω, the edge joining the vertex $\overline{u}$ that belongs to the solo D-pair to the vertex x outside S has weight 1/2, contradicting Claim J(b). Hence there is no edge joining v to a vertex in D, and so $g(S_v) = f(S_v)$. Thus either there is an edge joining v to some other vertex in B or there are two edges joining v to vertices in $V \setminus S$.

Since v has an S-private neighbor, $f(v) \geq 1$. By Claim A, there are two edges that join $\overline{v}$ to $V \setminus S$. By Claim D(d), each such edge has weight at least 1/6, and so $f(\overline{v}) \geq 1/3$.

Thus, $g(S_v) = f(S_v) = f(v) + f(\overline{v}) \geq 4/3$. By Claim J(a), $g(S_v) = 4/3$. Hence, $f(v) = 1$ and $f(\overline{v}) = 1/3$. Thus both edges that join the vertex $\overline{v}$ in C to vertices outside S have weight exactly $1/6$. By Claim D(c) and Claim L, each such edge joins $\overline{v}$ to a Z-vertex. Thus we have shown that every vertex in C is adjacent to two Z-vertices. By definition, every Z-vertex is triply dominated by S.

If there are two edges joining v to vertices in $V \setminus S$, then the edge joining v to its S-private neighbor has weight 1 and the other edge has weight 0. By definition of the function ω, this edge of weight 0 joins v to an X-vertex. This contradicts Claim M. Hence there is an edge joining v to some other vertex $w \in B$. Consider now the set $S' = S \setminus \{\overline{v}, \overline{w}\}$. Since the vertices $\overline{v}$ and $\overline{w}$ belong to C, their neighbors in $V \setminus S$ are triply dominated by S. Thus, the set S' is a smaller PDS of G (with v and w paired), a contradiction. QED

By Claim K and Claim N, the only pairs of vertices in S are D-pairs. In particular, there is no Z-vertex. The following result is a restatement of Claim J(b).

Claim O *Every vertex in $V \setminus S$ that is joined to a vertex that belongs to a solo D-pair is triply dominated by S.*

Claim P *There is no linked D-pair.*

PROOF. Suppose that S_v is a linked D-pair. By definition, there is an edge that joins S_v to some other D-pair, say $S_u = \{u, \overline{u}\}$. We may assume that $uv \in E$. We will refer to the edge joining vertices from distinct linked D-pairs as a *link edge*. By Claim J(c) there is a vertex $y \in V \setminus S$ whose neighbors in S are precisely $\overline{u}$ and $\overline{v}$. Further, $\omega(y\overline{u}) = \omega(y\overline{v}) = 1/2$.

Let r, s, x and z be defined as in the proof of Claim H. Thus, s is the neighbor of u different from v and $\overline{u}$, and r is the neighbor of $\overline{u}$ different from u and y. Further, x is the neighbor of $\overline{v}$ different from v and y, and z is the neighbor of v different from u and $\overline{v}$.

If $\omega(\overline{v}x) < 1/2$ and $\omega(vz) < 1/2$, then as in the proof of Claim H, the vertex x must be a Z-vertex, contradicting their nonexistence. Hence, $\omega(\overline{v}x) \geq 1/2$ or $\omega(vz) \geq 1/2$. Since $\omega(\overline{v}x) \geq 1/3$ and $\omega(vz) \geq 1/3$, we have that $\omega(\overline{v}x) + \omega(vz) \geq 1/2 + 1/3 = 5/6$. Thus, $g(S_v) = f(S_v) = \omega(vz) + \omega(\overline{v}x) + \omega(\overline{v}y) \geq 5/6 + 1/2 = 4/3$. By Claim J(a), $g(S_v) = 4/3$. Hence either $\omega(\overline{v}x) = 1/2$ and $\omega(vz) = 1/3$ or $\omega(\overline{v}x) = 1/3$ and $\omega(vz) = 1/2$. This is true for every linked D-pair S_v. The three possible weights of edges incident with vertices in $S_u \cup S_v$, up to symmetry, are illustrated in Figure 6.

Claim P(a) *We may assume there is no linked D-pair where both u and v are incident with an edge of weight $1/3$, as illustrated in Figure 6(b).*

PROOF. Suppose that both u and v are incident with an edge of weight $1/3$, as illustrated in Figure 6(b). That is, $\omega(us) = 1/3$, and so both s and z are triply dominated by S (possibly, $s = z$). Further, $\{s, z\} \cap \{x, r\} = \emptyset$. Let w be a neighbor of s not in $S_u \cup S_v$. Let

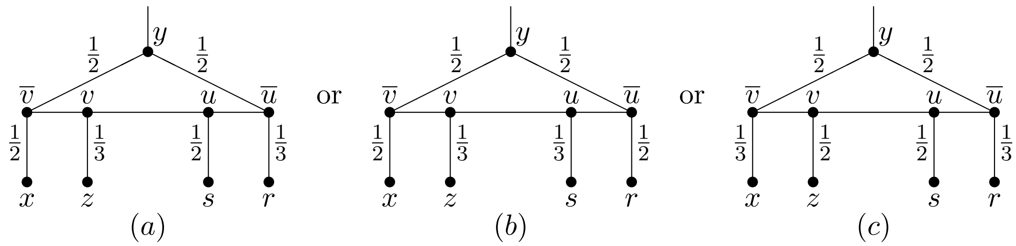

Figure 6: The possible weights of edges incident with vertices in $S_u \cup S_v$.

$S' = (S \setminus \{\overline{u}, u, v, \overline{w}\}) \cup \{s, y\}$. If S' dominates V, then S' would be a smaller PDS of G (with $\overline{v}$ and y paired and s and w paired), a contradiction. Hence, S' does not dominate V. It follows that there exists a vertex outside S adjacent only to $\overline{u}$ and $\overline{w}$; necessarily this is the vertex r. Since the vertex r is doubly dominated by S and is adjacent to $\overline{w}$, we remark that S_w is a linked D-pair.

If $s \neq z$, then s has a neighbor p, distinct from w, not in $S_u \cup S_v$. But then $S' = (S \setminus \{\overline{u}, u, v, \overline{p}\}) \cup \{s, y\}$ would be a smaller PDS of G (with $\overline{v}$ and y paired and s and p paired), a contradiction. Hence, $s = z$.

We now let $S' = (S \setminus \{u, v, \overline{v}, \overline{w}\}) \cup \{s, y\}$. If S' dominates V, then S' would be a smaller PDS of G (with $\overline{u}$ and y paired and s and w paired), a contradiction. Hence, S' does not dominate V. It follows that there exists a vertex outside adjacent only to $\overline{v}$ and $\overline{w}$; this is necessarily the vertex x, which is doubly dominated by S. Thus $\overline{w}$ is incident with two edges both of weight $1/2$. The edges incident with vertices in $S_u \cup S_v$ are illustrated in Figure 7.

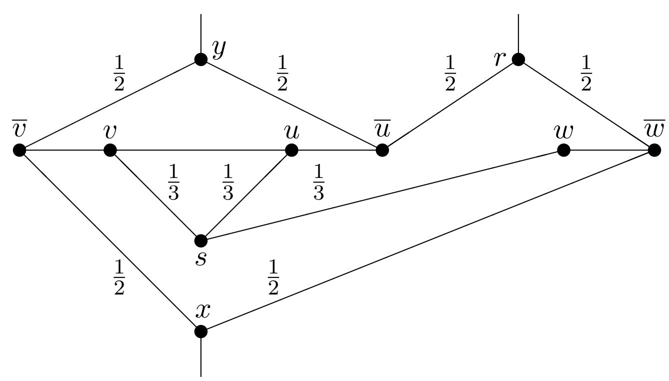

Figure 7: The edges incident with vertices in $S_u \cup S_v$.

Now, define a vertex outside S as a *bonding vertex* if it is adjacent to exactly two vertices of S and these neighbors belong to linked D-pairs that are joined by a link edge. Examination of Figure 6 shows that every vertex in a linked D-pair incident with two edges of weight $1/2$ is adjacent to a bonding vertex. It follows that $\overline{w}$ should be adjacent to a bonding vertex. However, its two neighbors outside S are r and x, but neither is a bonding vertex, a contradiction. QED

Claim P(b) *There is a linked D-pair where exactly one of u and v is incident with an edge of weight $1/2$, as illustrated in Figure 6(a).*

PROOF. Suppose that both ends of every link edge are incident with an edge of weight $1/2$, as illustrated in Figure 6(c). Let S consist of $2a$ linked D-pairs and b solo D-pairs. Thus, $|S| = 4a + 2b$. Let W be the set of all vertices in $V \setminus S$ that are joined to S by exactly two edges. Then, $|W| = 2a$ (each linked D-pair is incident with two edges to W) and $G[W]$ is regular of degree 1. Since every vertex in a linked D-pair is incident with an edge of weight $1/2$, the set W dominates the set of all vertices that belong to a linked D-pair. Thus, if S^b denotes the set of vertices in solo D-pairs, then $S' = S^b \cup W$ is a PDS of G with $|S'| = 2a + 2b$, a contradiction. QED

By Claims P(a) and (b), we may assume that the weights of the edges incident with vertices in $S_u \cup S_v$ are as illustrated in Figure 6(a).

Suppose that $x = s$. Then, u and $\overline{v}$ have a common neighbor which is doubly dominated by S. Since the vertex r is triply dominated, there is a neighbor w of r not in $S_u \cup S_v$. Let $S' = (S \setminus \{u, \overline{u}, v, \overline{w}\}) \cup \{x, r\}$. Then, S' is a smaller PDS of G (with x and $\overline{v}$ paired and r and w paired), a contradiction. Hence, $x \neq s$. Since x is doubly dominated by S and $x \neq s$, there is a neighbor p of x not in $S_u \cup S_v$. Let $S' = (S \setminus \{v, \overline{v}, \overline{p}\}) \cup \{x\}$. Then, S' is a smaller PDS of G (with x and p paired), a contradiction. QED

By Claim K, Claim N and Claim P, the only pairs of vertices in S are solo D-pairs. Thus every vertex of S has two neighbors outside S. By Claim O, every vertex outside S is triply dominated by S. Further, every edge in $[S, V \setminus S]$ has weight $1/3$.

Claim Q *There is no D-pair with a common neighbor.*

PROOF. Assume there is a (solo) D-pair $S_v = \{v, \overline{v}\}$ with a common neighbor a. Since the vertex a is triply dominated, there is a neighbor, u say, of a not in S_v. Let $S' = (S \setminus \{\overline{u}, v, \overline{v}\}) \cup \{a\}$. If S' dominates V, then S' would be a smaller PDS of G (with a and u paired), a contradiction. Hence, S' does not dominate V. Since every vertex outside S is triply dominated, there exists a vertex b adjacent to all of $\overline{u}$, v and $\overline{v}$. Let c be the neighbor of u in $V \setminus S$ different from a. Since the vertex c is triply dominated, there is a neighbor, w say, of c not in S_u. (The edges incident with vertices in S_v are illustrated in Figure 8.) But then the set $S' = (S \setminus \{u, \overline{u}, \overline{v}, \overline{w}\}) \cup \{b, c\}$ is a smaller PDS of G (with b and v paired and c and w paired), a contradiction. QED

We now return to our main proof. By Claim Q, no D-pair has a common neighbor. Thus every vertex outside S has three neighbors in S and these neighbors belongs to distinct D-pairs. Let S_v be a D-pair. Let a and b be the two neighbors of v outside S and let c and d be the two neighbors of $\overline{v}$ outside S. By Claim Q, these four neighbors of S_v are distinct. Let u be a neighbor of a different from v. Since every vertex outside S has neighbors that

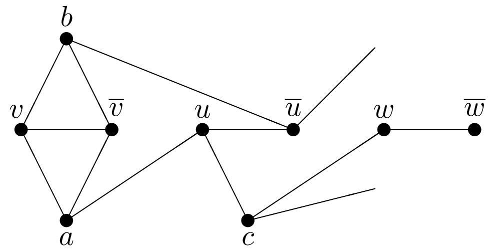

Figure 8: A D-pair S_v with a common neighbor.

belongs to three distinct D-pairs, the vertex c has a neighbor w that does not belong to $S_u \cup S_v$.

Let $S_1 = (S \setminus \{\overline{u}, v, \overline{v}, \overline{w}\}) \cup \{a, c\}$. If S_1 dominates V, then S_1 would be a smaller PDS of G (with a and u paired and c and w paired), a contradiction. Hence, S_1 does not dominate V. Therefore there must be a vertex outside S whose three neighbors are $\overline{u}$, $\overline{w}$, and exactly one of v and $\overline{v}$. Hence such a vertex must be b or d. We may assume that b is adjacent to both $\overline{u}$ and $\overline{w}$. Hence the graph shown in Figure 9 is a subgraph of G.

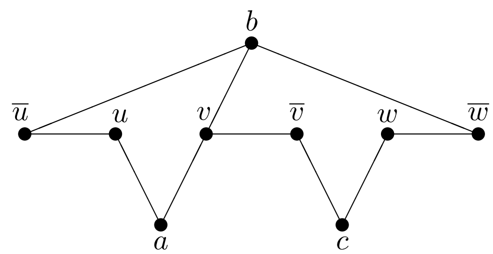

Figure 9: A subgraph of G.

Let $S_2 = (S \setminus \{u, v, \overline{v}, \overline{w}\}) \cup \{b, c\}$. If S_2 dominates V, then S_2 would be a smaller PDS of G (with b and $\overline{u}$ paired and c and w paired), a contradiction. Hence, S_2 does not dominate V. Therefore there must be a vertex outside S whose three neighbors are u, $\overline{w}$, and exactly one of v and $\overline{v}$. Such a vertex is therefore either a or d. We show that it must be d.

Claim R *The vertex a is not adjacent to $\overline{w}$.*

PROOF. Suppose that vertex a is adjacent to $\overline{w}$. Then, a and b have two common neighbors, namely v and $\overline{w}$. Let p be a neighbor of c not in $S_v \cup S_w$. Possibly, $S_p = S_u$. At least one of a and b, say the former, is not adjacent to p. Let $S_3 = (S \setminus \{\overline{v}, w, \overline{w}, \overline{p}\}) \cup \{a, c\}$. If S_3 dominates V, then S_3 would be a smaller PDS of G (with a and v paired and c and p paired), a contradiction. Hence, S_3 does not dominate V. Therefore the three neighbors of the vertex d are $\overline{p}$, $\overline{v}$ and w.

Suppose $S_p = S_u$. Then, $u = \overline{p}$ and $\overline{u} = p$. Thus, G is the graph shown in Figure 10. But then $S' = \{a, c, v, w\}$ is a smaller PDS of G (with a and v paired and c and w paired), a contradiction. Hence, $S_p \neq S_u$.

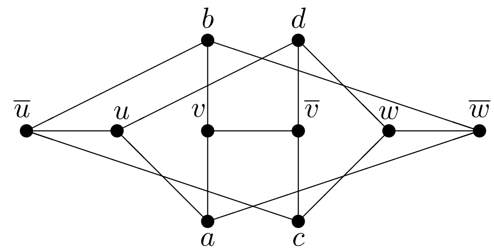

Figure 10: The Graph G.

Thus the graph shown in Figure 11 is a subgraph of G. Let e be the neighbor of u outside S different from a, and let f be the neighbor of $\overline{u}$ outside S different from b. Let q be a neighbor of e different from u. Since every vertex outside S has neighbors that belong to three distinct D-pairs, the vertex f has a neighbor r that does not belong to $S_u \cup S_q$. Note that $r \notin S_v \cup S_w$.

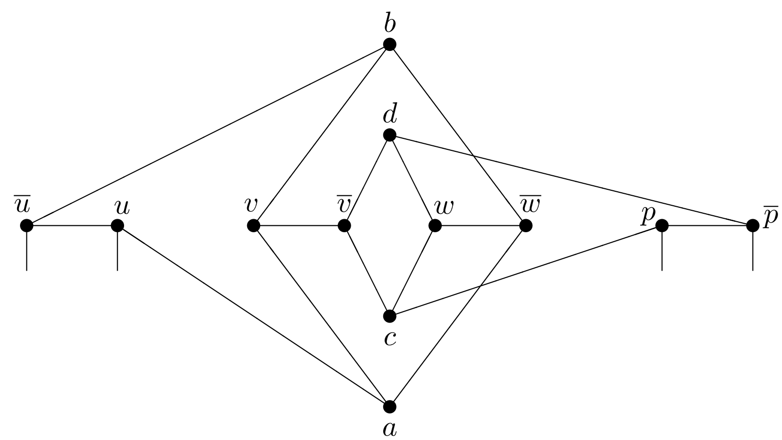

Figure 11: A subgraph of G.

Let $S_4 = (S \backslash \{u, \overline{u}, \overline{q}, \overline{r}\}) \cup \{e, f\}$. If S_4 dominates V, then S_4 would be a smaller PDS of G (with e and q paired and f and r paired), a contradiction. Hence, S_4 does not dominate V. Therefore there must be a vertex outside S whose three neighbors are $\overline{q}$, $\overline{r}$, and exactly one of u and $\overline{u}$. But neither a nor b is adjacent to a vertex in $S_q \cup S_r$, a contradiction. QED

We now return to our main proof one last time. By Claim R, the vertex outside S whose three neighbors are u, $\overline{w}$, and exactly one of v and $\overline{v}$, must be the vertex d which is the neighbor of $\overline{v}$ outside S different from c.

Let $S_3 = (S \backslash \{u, v, \overline{v}, w\}) \cup \{b, d\}$. If S_3 dominates V, then S_3 would be a smaller PDS of G (with b and $\overline{u}$ paired and d and $\overline{w}$ paired), a contradiction. Hence, S_3 does not dominate V. Thus the vertex a must be adjacent to w. We now let $S_4 = (S \setminus \{\overline{u}, v, \overline{v}, w\}) \cup \{a, d\}$. If S_4 dominates V, then S_4 would be a smaller PDS of G (with a and u paired and d and $\overline{w}$ paired), a contradiction. Hence, S_4 does not dominate V. Thus the vertex c must be adjacent to $\overline{u}$, implying that G is the Petersen graph shown in Figure 12, as desired. This completes the proof of Theorem 5. QED

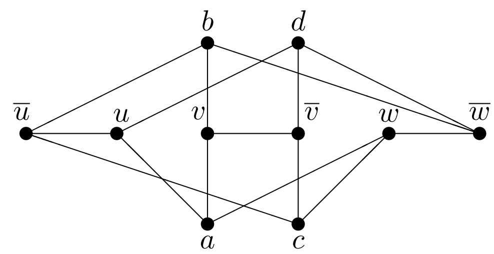

Figure 12: The Petersen Graph.

3 Concluding Remarks

Chen, Sun and Xing [1] posed the following conjecture.

Conjecture 1 ([1]) *If G is a connected graph of order $n \geq 11$ with minimum degree at least 3, then $\gamma_{\rm pr}(G) \leq 4n/7$.*

We remark that the sharpness of the bound in Conjecture 1, if true, is not discussed in [1]. We close with the slightly stronger conjecture that every connected graph with minimum degree at least three, with the exception of the Petersen graph P shown in Figure 1, has paired-domination number at most four-sevenths its order.

Conjecture 2 *If $G \neq P$ is a connected graph of order n with $\delta(G) \geq 3$, then $\gamma_{\rm pr}(G) \leq 4n/7$.*

If Conjecture 2 is true, then the bound is achieved, for example, by the graph F shown in Figure 13.

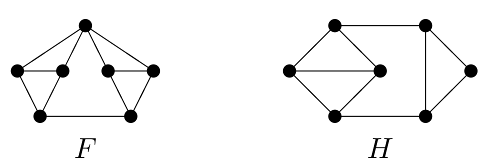

Figure 13: A graphs F and H.

We remark that although there is no known infinite family of graphs which achieves the upper bound of Conjecture 2, there is an infinite family of connected graphs of order n with $\delta(G) \geq 3$ and with $\gamma_{\rm pr}(G)$ approaching $4n/7$ for large n. For example, let $\mathcal{G}$ be the family of all graphs G that can be obtained from $k \geq 3$ disjoint copies of the graph H shown in Figure 13 by adding a new vertex and joining it to the vertex of degree 2 in each of the k copies of H. A graph G in the family $\mathcal{G}$ is illustrated in Figure 14. Such a graph G has order $n = 7k + 1$ and $\gamma_{\rm pr}(G) = 4k = 4(n-1)/7$.

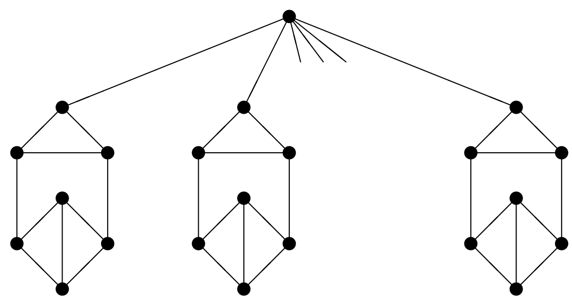

Figure 14: A graph G in the family $\mathcal{G}$.

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